

Model Categories - Self Study

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Abstract

This text contains my notes on model category theory. These notes were synthesized from a combination of Mark Hovey's *Model Categories*, and Emily Riehl's *Categorical Homotopy Theory*. The sections of my notes on Riehl's text can also be found in my full notes for her text; I made these notes since I am using both texts simultaneously, so it felt wrong to only include these notes in my notes on her book.

I wrote these notes during the summer between my Sophomore and Junior years. My goal was to obtain rigorous understanding of basic model category theory, and to explore Joyal's model structure on $\mathbf{sSet}$, such that I could begin exploring quasi-categories.

Chapter 1

Model Categories (Hovey 1)

We begin with an initial exploration of model categories following Hovey's more classical perspective, which builds atop that of Quillen. What Quillen originally described as *closed* model categories have been determined to be the more important class of examples, so they have come to be known as model categories in the modern literature. The goal is to provide for a homotopical category some extra structure, in the form of specified classes of morphisms known as fibrations and cofibrations, to make the homotopy category better behaved and less difficult to work with.

Conventions vary mildly on model categorical definitions. Hovey states that we will require functorial factorizations be part of the structure of a model category, which more classical texts did not. I do not yet know Riehl's perspective on this, but either way this distinction, and others like it, are immaterial since in practice all relevant model categories satisfy all definitions.

1.1 The definition of a model category

Given a category $\mathcal{C}$, define by $\text{Map } \mathcal{C}$ the category whose objects are morphisms in $\mathcal{C}$, and whose morphisms are commutative squares; meaning for objects $f : a \rightarrow b$ and $g : c \rightarrow d$, a morphism $f \rightarrow g$ is a pair (h, k) such that

$$\begin{array}{ccc} a & \xrightarrow{h} & c \\ f \downarrow & & \downarrow g \\ b & \xrightarrow{k} & d \end{array}$$

commutes. One also notes that this is equivalent to the comma category $\text{Map } \mathcal{C} = (\mathcal{C} \downarrow \mathcal{C})$. The operations of domain and codomain promote to functors $\text{dom}, \text{cod} : \text{Map } \mathcal{C} \rightarrow \mathcal{C}$, defined by $f \mapsto \text{dom } f$ and $(h, k) \mapsto h$; this works because $h : \text{dom } f \rightarrow \text{dom } g$. Codomain is defined similarly. The fact that this is functorial is a subtle yet important nuance in the following definition:

Definition 1.1.1 (Retract/Functorial Factorization). Let $\mathcal{C}$ be a category. A map $f : a \rightarrow b$ in $\mathcal{C}$ is a *retract* of a map $g : c \rightarrow d$ in $\mathcal{C}$ if f is a retract of g as objects of $\text{Map } \mathcal{C}$. More precisely, f is a retract of g if there is a diagram

$$\begin{array}{ccccc} & & 1 & & \\ & \searrow & & \nearrow & \\ a & \xrightarrow{\quad} & c & \xrightarrow{\quad} & a \\ f \downarrow & & g \downarrow & & \downarrow f \\ b & \xrightarrow{\quad} & d & \xrightarrow{\quad} & b \\ & \nwarrow & & \swarrow & \\ & & 1 & & \end{array}$$

A *functorial factorization* is a pair (α, β) of endofunctors $\text{Map } \mathcal{C} \rightarrow \text{Map } \mathcal{C}$ such that $f = \beta(f) \circ \alpha(f)$ for all $f \in \text{Map } \mathcal{C}$, and also such that

- $\text{dom} \circ \alpha = \text{dom}$;
- $\text{cod} \circ \alpha = \text{dom} \circ \beta$;
- $\text{cod} \circ \beta = \text{cod}$.



Remark 1.1.2. Further commentary on the definition of a functorial factorization is beneficial. Note first that the composition $\beta(f) \circ \alpha(f)$ is composition in $\mathcal{C}$, since f is an object of $\text{Map } \mathcal{C}$. We thus obtain the following remarks from this definition:

1. Every arrow f admits a factorization into two components, denoted as $\beta(f)$ and $\alpha(f)$;
2. If one only considers the action of dom and cod on objects, the information in the three bullet points is not anything more powerful than the condition that $\beta(f) \circ \alpha(f)$ be well-defined. However, when one considers the action on morphisms, we obtain that for a morphism $(h, k) : f \rightarrow g$ in $\text{Map } \mathcal{C}$,

$$\alpha(k) = (\text{cod} \circ \alpha)(h, k) = (\text{dom} \circ \beta)(h, k) = \beta(h).$$

Here I use a minor abuse of notation and denote by $\alpha(k)$ the arrow $(\text{dom} \circ \alpha)(h, k)$ that replaces k in the square diagram, and similarly for $\beta(h)$. We also obtain that $\alpha(h) = h$ and $\beta(k) = k$. Thus the diagrams

$$\begin{array}{ccc} a & \xrightarrow{h} & c \\ \alpha(f) \downarrow & & \downarrow \alpha(g) \\ b & \xrightarrow{\alpha(k)} & d \end{array} \quad \begin{array}{ccc} a & \xrightarrow{\beta(h)} & c \\ \beta(f) \downarrow & & \downarrow \beta(g) \\ b & \xrightarrow{k} & d \end{array}$$

glue together to form the diagram

$$\begin{array}{ccc} & a & \xrightarrow{h} & b \\ & \downarrow \alpha(f) & & \downarrow \alpha(g) \\ f \swarrow & e_f & \xrightarrow{\alpha(k)=\beta(f)} & e_g \searrow g \\ & \downarrow \beta(f) & & \downarrow \beta(g) \\ & b & \xrightarrow{k} & d \end{array}$$

3. By the above diagram, functorial factorizations preserve retracts;
4. Note also that if

$$f \xrightarrow{\phi} g \xrightarrow{\psi} f$$

is a retract, then $\psi \circ \phi = 1$, and thus ϕ is mono and ψ is epi.

Also note that the definition as given in Hovey does not make precise that domain and codomain should be viewed as functors; this is clarified in an errata that can be found [on the nLab](#).

Definition 1.1.3 (Lifting Property). Let $i : a \rightarrow b$ and $p : x \rightarrow y$ be maps in $\mathcal{C}$. Then i has the *left lifting property with respect to p* and p has the *right lifting property with respect to i* if, for every commutative diagram of solid arrows

$$\begin{array}{ccc} a & \xrightarrow{f} & x \\ i \downarrow & \nearrow h & \downarrow p \\ b & \xrightarrow{g} & y \end{array}$$

there exists a dotted arrow as indicated rendering the diagram commutative. ♥

Remark 1.1.4. Note that the diagram of solid arrows is the datum of a morphism $(f, g) : i \rightarrow p$ in $\text{Map } \mathcal{C}$; a lift h can then be viewed as an arrow in $\mathcal{C}$ such that $(f, h) : i \rightarrow 1_x$ and $(h, g) : 1_b \rightarrow p$.

Definition 1.1.5 (Model Structure). A *model structure* on a category $\mathcal{C}$ is three *subcategories* of $\mathcal{C}$ called the weak equivalences, fibrations, and cofibrations, together with two functorial factorizations (α, β) and (γ, δ) satisfying the following properties:

1. (2-out-of-3) If f, g are composable arrows in $\mathcal{C}$ such that two of f , g , and gf are weak equivalences, then so is the third;
2. (Retracts) If f, g are morphisms in $\mathcal{C}$ such that f is a retract of g and g is either weak equivalence, fibration, or cofibration, then so is f ;
3. (Lifting) A map is an *trivial cofibration* if it is both a cofibration and a weak equivalence, and the same for *trivial fibrations*. Then trivial cofibrations have the left lifting property with respect to fibrations, and cofibrations have the left lifting property with respect to trivial fibrations;
4. (Factorization) For any morphism f , $\alpha(f)$ is a cofibration, $\beta(f)$ is a trivial fibration, $\gamma(f)$ is a trivial cofibration, and $\delta(f)$ is a fibration.

♥

Definition 1.1.6 (Model Category). A *model category* $\mathcal{C}$ is a complete and cocomplete category with a specified model structure. ♥

Remark 1.1.7. For the naive reader, note that referring to maps as *weak equivalences*, *fibrations*, or *cofibrations* simply means that those maps are in the relevant subcategory of $\mathcal{C}$.

Of course, the intuition for weak equivalences is most easily viewed via homotopy equivalences. Given homotopy equivalences f and g , one immediately has that gf is a homotopy equivalence; conversely given gf and f homotopy equivalences, one may construct a homotopy inverse for g making g a homotopy equivalence, and the same for f . As we well know, the idea of formally inverting a specified class of morphisms is very prominent in mathematics, with the inversion of homotopy equivalences in Top closely tied to quotienting by homotopy, and the inversion of quasi-isomorphisms in $\text{Ch}(\mathcal{A})$ giving rise to the derived category $\mathcal{D}(\mathcal{A})$, a central topic in homological algebra. Due to my lacking background in algebraic topology, I lack the natural intuition for what fibrations and cofibrations should be, which gives me an opportunity to develop a uniquely categorical intuition before I end up reading on algebraic topology.

With the 2-out-of-3 property thoroughly motivated, we give some more detailed explanation for the other properties.

- (Retracts) This property simply says that fibrations, cofibrations, and weak equivalences are

closed under retracts.

- (Lifting) It is not detailed thoroughly in this text, but for a collection $\mathcal{L}$ of maps to have the left lifting property against another collection $\mathcal{R}$ of maps, we must have that *every* map in $\mathcal{L}$ has the left lifting property with respect to *every* map in $\mathcal{R}$.
- (Factorization) This property says that each map admits a factorization into a trivial cofibration and a fibration, and a cofibration and a trivial fibration, in the sense of definition 1.1.1 and remark 1.1.2.

Remark 1.1.8. Emily Riehl in *Categorical Homotopy Theory* requires that model categories satisfy a stronger 2-of-6 property, instead of the 2-out-of-3 property. It's proven in her text that in the case of model categories, these properties become equivalent; I will prove this when I read her discussion of model categories. The more important distinction she makes in her text is that the class of weak equivalences needs to be *wide*, meaning it contains all objects. This is a simple way of saying it contains all identities, and thus by a very simple proof it contains all isomorphisms. Hovey does not specify this, but it would still make sense for a model category's class of weak equivalences to contain all isomorphisms. This is reconciled by a result we will not see until later. The lifting axiom will later be shown to *characterize* trivial (co)fibrations, meaning we will show maps are trivial cofibrations *if and only if* they have the left lifting property with respect to fibrations, and similarly for the dual statement. Note that isomorphisms admit a lift for any diagram: let f and g be inverses. Then

$$\begin{array}{ccc}
 x & \xrightarrow{h} & a \\
 f \downarrow & \nearrow h & \downarrow p \\
 & x & \\
 g \nearrow & & \\
 y & \xrightarrow{k} & b
 \end{array}$$

commutes, since $hgf = h$ gives the commutivity of the top triangle, and since the outer square commutes by hypothesis, $kf = ph \implies k = phg$ as required for the bottom triangle. Thus f has the left lifting property against all arrows; in particular fibrations. Thus f , as will be shown in lemma 1.1.18, must be a trivial cofibration, and thus it is by definition a weak equivalence.

We assume small limits and colimits exist largely for convenience, since most model categories in practice admit all small limits. This differs from Quillen's original requirement that only finite limits exist. We also assume the stronger convention that factorizations be functorial, since once again they generally are in practice. We also assume the functorial factorizations are *part of the structure*, which is a subtle difference and ensures that some maps will be natural going forward. We always leave the model structure implicit and refer to a model category $\mathcal{C}$, because oh my god can you imagine having to write out like $(\mathcal{C}, (\alpha, \beta), (\gamma, \delta), \mathcal{F}, \mathcal{Q}, \mathcal{W})$ every time you wanted to refer to a model category? That would be so aids lmfao

Example 1.1.9. Any complete and cocomplete category $\mathcal{C}$ can be given three different model structures by choosing one of the subcategories to be isomorphisms, and the other two subcategories to be all of $\mathcal{C}$. The only nontrivial things to be shown are that

1. retracts of isomorphisms are isomorphisms;
2. the lifting axiom holds;
3. functorial factorizations admit a definition.

Proving (1) is not incredibly difficult. Let $g : x \rightarrow y$ be an isomorphism, and let $f : a \rightarrow b$ be a

retract of g :

$$\begin{array}{ccccc} a & \xrightarrow{h} & x & \xrightarrow{h'} & a \\ f \downarrow & & g \downarrow & & \downarrow f \\ b & \xrightarrow{k} & y & \xrightarrow{k'} & b \end{array}$$

By commutativity of the diagram, we have that $k'g = fh'$, and thus $k' = fh'g^{-1}$. We thus construct the diagram

$$\begin{array}{ccccc} a & \xrightarrow{h} & x & \xrightarrow{h'} & a \\ f \downarrow & & g \downarrow & & \downarrow f \\ b & \xrightarrow{k} & y & \xrightarrow{k'} & b \\ & \searrow g^{-1}k & \downarrow g^{-1} & \nearrow fh' & \\ & & x & & \end{array}$$

Since the composition along the lower row is the identity, we obtain that $fh'g^{-1}k = 1$, and thus $h'g^{-1}k$ is a right inverse for f . Showing that it is a left inverse is more simple; by commutivity of the left square we have

$$gh = kf \implies h = g^{-1}kf \implies 1 = h'g^{-1}kf,$$

where we have used the fact that $h'h = 1$. Thus, we have constructed a two-sided inverse for f , and f must be an isomorphism.

Now we prove (2) for each case. We will show extremely soon that model categories are dualizable, so we need only show it for one of the cofibrations or fibrations.

1. Start by taking weak equivalences to be precisely isomorphisms. Thus trivial cofibrations and trivial fibrations are precisely isomorphisms, which we showed in remark 1.1.8 to have the left lifting property with respect to all maps; the same proof shows they have the right lifting property as well (with notation as in the problem, the diagonal arrow is gk). Thus trivial (co)fibrations have the left *and* right lifting properties with respect to all maps, and thus (co)fibrations. This proves the statement.
2. Now take cofibrations to be isomorphisms. All arrows are trivial fibrations, and all cofibrations are trivial cofibrations. Since trivial cofibrations are in particular isomorphisms, they have the left lifting property with respect to all maps, and thus fibrations. Note also that all cofibrations are isomorphisms, and thus have the left lifting property with respect to all maps, in particular trivial fibrations.

Now we prove (3). We once again work case-by-case.

1. Take weak equivalences to be isomorphisms. Then one defines $\alpha(f) = f$ and $\beta(f) = 1_{\text{cod } f}$, similarly $\gamma(f) = 1_{\text{dom } f}$ and $\delta(f) = f$. The functorial factorization condition forces the action on a morphism $(h, k) : f \rightarrow g$ to satisfy

$$(\text{dom} \circ \alpha)(h, k) = (\text{dom} \circ \gamma)(h, k) = h, \quad (\text{cod} \circ \beta)(h, k) = (\text{cod} \circ \delta)(h, k) = k.$$

One defines $(\text{cod} \circ \alpha)(h, k) = k$ and $(\text{cod} \circ \gamma)(h, k) = h$, and the corresponding definition for β, δ . This is indeed functorial, since all the factorization (α, β) does is attach a square to the

bottom:

$$\begin{array}{ccc}
 x & \xrightarrow{h} & a \\
 f \downarrow & & \downarrow g \\
 y & \xrightarrow{k} & b \\
 1 \downarrow & & \downarrow 1 \\
 y & \xrightarrow{k} & b
 \end{array}$$

and similarly for (γ, δ) . Note that in this case, we find that α, δ are the identity and β, γ can be described as the identity on the codomain.

2. I will not go into as much detail for this case since the proof is basically the same, but with some things switched around. Also because this stuff is kinda trivial. Let isomorphisms be cofibrations. Since identities are cofibrations, we decompose every map as itself composed with 1; this produces both functorial factorizations, since all cofibrations are trivial and all fibrations are trivial.

Example 1.1.10 (Product Model Structure). Let $\mathcal{C}$ and $\mathcal{D}$ be model categories. One defines the *product model structure* on $\mathcal{C} \times \mathcal{D}$ by taking maps (f, g) to be cofibrations (respectively fibrations, weak equivalences) if and only if both f and g are cofibrations (respectively fibrations, weak equivalences). The verification of all of the axioms is kind of trivial, since everything just decomposes into noting that the axioms hold in both categories, so it holds for the pairs by definition of the product category. The functorial factorizations are defined as the product; for example

$$\alpha_{\mathcal{C}} \times \alpha_{\mathcal{D}} : \text{Map } \mathcal{C} \times \text{Map } \mathcal{D} \rightarrow \text{Map } \mathcal{C} \times \text{Map } \mathcal{D},$$

where we note that $\text{Map } \mathcal{C} \times \text{Map } \mathcal{D} \cong \text{Map}(\mathcal{C} \times \mathcal{D})$.

Remark 1.1.11. Model category axioms are self-dual, meaning if $\mathcal{C}$ is a model category, $\mathcal{C}^{\text{op}}$ inherits a corresponding model structure by taking cofibrations in $\mathcal{C}^{\text{op}}$ to be fibrations in $\mathcal{C}$; fibrations in $\mathcal{C}^{\text{op}}$ to be cofibrations in $\mathcal{C}$; and weak equivalences stay the same. The functorial factorizations are inverted: denote by $(-)'$ the corresponding functor for $\mathcal{C}^{\text{op}}$. Then

- $\alpha' = \delta^{\text{op}}$;
- $\beta' = \gamma^{\text{op}}$;
- $\gamma' = \beta^{\text{op}}$;
- $\delta' = \alpha^{\text{op}}$.

We denote by $D\mathcal{C}$ the opposite model category of $\mathcal{C}$ with this model structure, and call it the *dual model category* of $\mathcal{C}$. Note that $D^2\mathcal{C} = \mathcal{C}$. Of course this means we need only prove a statement about cofibrations to prove the dual for fibrations, and vice versa.

Terminology 1.1.12. Since model categories are complete and cocomplete, they have initial and terminal objects. We denote by 0 the initial object, and 1 or $*$ the terminal object. An object $x \in \mathcal{C}$ is *cofibrant* if the arrow $0 \rightarrow x$ is a cofibration, and is called *fibrant* if the arrow $x \rightarrow *$ is a fibration. A model category is *pointed* when the map $0 \rightarrow *$ is an isomorphism.

Construction 1.1.13. Let $\mathcal{C}$ be a model category. Define by $\mathcal{C}_*$ the category $(* \downarrow \mathcal{C})$ under the terminal object. Recall this is the usual definition of a based category, as objects are arrows

$f : * \rightarrow x$, which are intuitively thought of as selecting a basepoint f for x . Recall morphisms are basepoint-preserving maps, so maps $\varphi : (x, f) \rightarrow (y, g)$ such that $g = \varphi \circ f$.

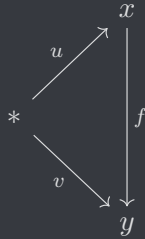
Note the obvious functor $(-)_+ : \mathcal{C} \rightarrow \mathcal{C}_*$ given by $x \mapsto x \amalg * =: x_+$, with the obvious basepoint. This is clearly an adjunction $(-)_+ \dashv U$ for $U : \mathcal{C}_* \rightarrow \mathcal{C}$ the forgetful functor, and thus $(-)_+$ preserves colimits. Thus, $\mathcal{C}_*$ is cocomplete. One also shows fairly easily that $\mathcal{C}_*$ is complete with limits in $\mathcal{C}_*$ the same as the underlying object in $\mathcal{C}$. One typically denotes by $x \vee y$ the coproduct in $\mathcal{C}_*$, analogous to the wedge product in $\mathcal{Top}_*$.

We construct a model structure on $\mathcal{C}_*$ by defining maps f in $\mathcal{C}_*$ to be cofibrations (fibrations, weak equivalences) precisely when the underlying function $U(f)$ is a cofibration (fibration, weak equivalence).

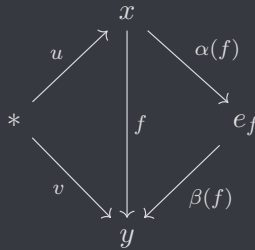
Proposition 1.1.14. *With definitions as in construction 1.1.13, $\mathcal{C}_*$ is a model category.*

Proof. This is really not a hard proof. Composition preserves the underlying function, so the 2-out-of-3 property holds. To show the subcategories are closed under retracts, we need only show that the underlying objects also form a retract. Indeed, this follows by functoriality of the forgetful functor, so this follows. We will now consider the factorization axiom, and will come back to lifting.

We may lift the functorial factorizations to $\mathcal{C}_*$ mostly for free: the functorial factorization condition guarantees the domain of $\alpha(h, k)$ is h , the codomain of $\beta(h, k)$ is k , and given a morphism $f \rightarrow g$ in $\text{Map } \mathcal{C}_*$ we would already have that these are basepoint preserving. The only work that needs to be done is to come up with a basepoint for e_f and e_g , the midpoint of $\beta(f) \circ \alpha(f)$, such that the maps are basepoint preserving. The correct definition manifests from the following diagrams: an arrow $f : (x, u) \rightarrow (y, v)$ in $\mathcal{C}_*$ can be viewed as a diagram



in $\mathcal{C}$. Since α and β are a factorization in $\mathcal{C}$, we obtain the diagram



in $\mathcal{C}$. Hence we may define the basepoint of e_f as $\alpha(f) \circ u : * \rightarrow e_f$, and it holds manifestly that the relevant diagram in $\mathcal{C}_*$ commutes, since it commutes in $\mathcal{C}$. All one has to do in order to realize this is to draw the diagram with and without the endpoints, look at it for 10 seconds, say “oh thats obvious”, and move on. The fact that these factor into cofibrations and trivial fibrations is

immediate by the definition of the model structure. The definitions of γ and δ follow in the same way.

Finally, we concern ourselves with the lifting properties. Suppose we have a lifting problem

$$\begin{array}{ccc} (x, u) & \xrightarrow{\mu} & (a, w) \\ p \downarrow & & \downarrow i \\ (y, v) & \xrightarrow{\nu} & (b, z) \end{array}$$

in $\mathcal{C}_*$, with p a cofibration and i a trivial fibration. Functoriality of U gives the lifting problem

$$\begin{array}{ccc} x & \xrightarrow{\mu} & a \\ p \downarrow & \nearrow h & \downarrow i \\ y & \xrightarrow{\nu} & b \end{array}$$

with the solution h as indicated, since $\mathcal{C}$ is a model category. Since the diagram without h commutes in $\mathcal{C}_*$, the outer arrows are basepoint preserving, meaning the diagram

$$\begin{array}{ccccc} & & w & & \\ & \nearrow & \curvearrowright & \searrow & \\ * & & x & \xrightarrow{\mu} & a \\ & \searrow & \downarrow p & \nearrow h & \downarrow i \\ & & y & \xrightarrow{\nu} & b \\ & \nearrow & \curvearrowright & \searrow & \\ & & z & & \end{array}$$

must also commute by basepoint preservation. To show that h solves the lifting problem in $\mathcal{C}_*$, it suffices to show it preserves basepoints, since we already know the diagram will commute. Thus, we must show $h\nu = w$. This follows immediately from commutivity of the diagram. The dual statements for trivial cofibrations and fibrations once again follow from remark 1.1.11. Thus, $\mathcal{C}_*$ is a model category. ■

Remark 1.1.15. The proof of proposition 1.1.14 did not rely on the universality of $*$ at all, meaning one can generate for any object $a \in \mathcal{C}$ a model category structure on $\mathcal{C}_a := (a \downarrow \mathcal{C})$ via the same assignments. Dualizing this gives us that the category $(\mathcal{C} \downarrow a)$ is also a model category.

Remark 1.1.16. Note that for any object $x \in \mathcal{C}$, the unique arrow $f : 0 \rightarrow x$ decomposes as

$$0 \xrightarrow{\alpha(f)} Q(x) \xrightarrow{\beta(f)} x,$$

where the reason for calling the in-between object $Q(x)$ will become clear soon. By definition 1.1.5, the arrow $\alpha(f)$ is a cofibration, so we have that the object $Q(x)$ is cofibrant. We thus obtain that the functor $Q = \text{cod} \circ \alpha$ maps objects x to *cofibrant objects* $Q(x)$. We call Q the *cofibrant replacement functor* of $\mathcal{C}$. We also obtain a natural transformation $q : Q \Rightarrow 1_{\mathcal{C}}$ with components given by $q_x = \beta(0 \rightarrow x)$, wherein each component is a trivial fibration. The fact that this is natural

comes from the fact that the functorial factorization defines, for each $f : x \rightarrow y$, a diagram

$$\begin{array}{ccccc} 0 & \xrightarrow{\alpha(0 \rightarrow x)} & Q(x) & \xrightarrow{\beta(0 \rightarrow x)} & x \\ \text{\textbf{I}} & & \downarrow \text{cod } \alpha(1, f) & & \downarrow f \\ 0 & \xrightarrow{\alpha(0 \rightarrow y)} & Q(y) & \xrightarrow{\beta(0 \rightarrow y)} & y \end{array}$$

First note that the outside square commutes since 0 is initial, and thus both inner squares commute since (α, β) is a functorial factorization. Note that $\text{cod } \alpha(1, f) = Q(f)$, so the right square expresses the naturality condition as desired. Dually, we have that $\text{dom } \delta$ is an *fibrant replacement*, usually denoted $R : \mathcal{C} \rightarrow \mathcal{C}$, and there is a natural trivial cofibration¹ $r : 1_{\mathcal{C}} \Rightarrow R$ with components given by $r_x = \gamma(0 \rightarrow x)$.

Lemma 1.1.17 (The Retract Argument). *Suppose one has the factorization $f = p \circ i$ in $\mathcal{C}$, and suppose that f has the left lifting property with respect to p . Then f is a retract of i . Dually, if f has the right lifting property with respect to i , then f is a retract of p .*

Proof. Define the domains via

$$\begin{array}{ccccc} & & f & & \\ & \nearrow & & \searrow & \\ x & \xrightarrow{i} & z & \xrightarrow{p} & y. \end{array}$$

Suppose f has the left lifting property with respect to p . Note that since $f = pi$, the following diagram of solid arrows

$$\begin{array}{ccc} x & \xrightarrow{i} & z \\ f \downarrow & \nearrow r & \downarrow p \\ y & \xrightarrow{\quad} & y \end{array}$$

commutes, and thus there is a dotted arrow r as indicated rendering the diagram commutative. This expresses the equality $i = rf$, and thus the diagram

$$\begin{array}{ccccc} x & \xrightarrow{\quad} & x & \xrightarrow{\quad} & x \\ f \downarrow & & \downarrow i & & \downarrow f \\ y & \xrightarrow{\quad} & z & \xrightarrow{p} & y \end{array}$$

commutes, since the equality expressed by the square on the right is simply $f = pi$. Note also by the diagram that $pr = 1$, and thus f is a retract of i . The other argument follows by duality. ■

Lemma 1.1.18. *Let $\mathcal{C}$ be a model category. Then a map is a (trivial) cofibration if and only if it has the left lifting property with respect to all fibrations (trivial fibrations). Dually, a map is a fibrant (a trivial fibrant) if and only if it has the right lifting property with respect to all trivial cofibrations (cofibrations).*

Proof. The model category axioms guarantee that (trivial) cofibrations lift against fibrations (trivial fibrations), so we need only show the converse. Let f have the left lifting property with

¹The terminology “natural [class of map]” means there is a natural transformation, and each component belongs to the specified type of map. These are also called “pointwise [class of map] natural transformations”.

respect to all trivial fibrations. Note that $f = \beta(f) \circ \alpha(f)$, and $\beta(f)$ is a trivial fibration. By assumption, f thus has the left lifting property with respect to $\beta(f)$, and by lemma 1.1.17 f is a retract of $\alpha(f)$. Since $\alpha(f)$ is a cofibration and cofibrations are closed under retracts, we have that f is a cofibration.

Next, let f have the left lifting property with respect to all fibrations, and note that $f = \delta(f) \circ \gamma(f)$, where $\delta(f)$ is a fibration. By assumption, f has the left lifting property with respect to $\delta(f)$, and by lemma 1.1.17 f is a retract of $\gamma(f)$. Since $\gamma(f)$ is a trivial cofibration, it is a weak equivalence and a cofibration, both of which are closed under retracts. Therefore, f is a trivial cofibration. The other statements follow by duality. $\blacksquare$

Remark 1.1.19. This suffices to prove remark 1.1.8, in particular that all isomorphisms are weak equivalences. In particular, isomorphisms are trivial cofibrations *and* trivial fibrations, since they have the retract argument against all maps (again, see remark 1.1.8).

Corollary 1.1.20. *Let $\mathcal{C}$ be a model category. Then cofibrations (trivial cofibrations) are closed under pushouts in the sense that if*

$$\begin{array}{ccc} a & \longrightarrow & c \\ f \downarrow & & \downarrow g \\ b & \longrightarrow & d \end{array}$$

is a pushout square and f is a cofibration (trivial cofibration), then so is g . Dually, fibrations are closed under pullbacks in the sense that if

$$\begin{array}{ccc} a & \longrightarrow & c \\ g \downarrow & & \downarrow f \\ b & \longrightarrow & d \end{array}$$

is a pullback and f is a fibration (trivial fibration), then so is g .

Proof. Consider the pushout case. Let

$$\begin{array}{ccc} a & \xrightarrow{\ell_1} & c \\ f \downarrow & & \downarrow g \\ b & \xrightarrow{k_1} & d \end{array}$$

be the pushout square, and consider a lifting problem

$$\begin{array}{ccc} c & \xrightarrow{\ell_2} & x \\ g \downarrow & & \downarrow h \\ d & \xrightarrow{k_2} & y. \end{array}$$

Attaching the pushout square gives the diagram

$$\begin{array}{ccccc} a & \xrightarrow{\ell_1} & c & \xrightarrow{\ell_2} & x \\ f \downarrow & & g \downarrow & & \downarrow h \\ b & \xrightarrow{k_1} & d & \xrightarrow{k_2} & y. \end{array}$$

If f has the left lifting property with respect to h , then there is a lift

$$\begin{array}{ccccc} a & \xrightarrow{\ell_1} & c & \xrightarrow{\ell_2} & x \\ f \downarrow & & & \nearrow r & \downarrow h \\ b & \xrightarrow{k_1} & d & \xrightarrow{k_2} & y. \end{array}$$

indicated by the dotted line. Commutivity of these diagrams gives the diagram of solid arrows

$$\begin{array}{ccccc} a & \xrightarrow{\ell_1} & c & & \\ f \downarrow & & g \downarrow & \searrow \ell_2 & \\ b & \xrightarrow{k_1} & d & \xrightarrow{r'} & x \\ & \searrow r & & & \end{array}$$

and by universality of pushout, there is a unique dashed arrow r' rendering the diagram commutative. We thus obtain that the diagram

$$\begin{array}{ccc} c & \xrightarrow{\ell_2} & x \\ g \downarrow & \nearrow r' & \\ d & & \end{array}$$

commutes. To show the full relevant diagram commutes, note that composing in the bottom right of the pushout diagram by h gives a unique arrow $hr'k_1 = hr$, and since $hr = k_1k_2$, we have $hr'k_1 = k_2k_1$. By uniqueness of the arrow hr' , we have that $hr' = k_2$, and thus we may fill in our diagram to obtain the commutative diagram

$$\begin{array}{ccccc} c & \xrightarrow{\ell_2} & x & & \\ g \downarrow & & \nearrow r' & & \downarrow h \\ d & \xrightarrow{k_2} & y & & \end{array}$$

Thus, if f has the left lifting property with respect to any arrow h , then so does g . The statement of the corollary then follows by remark 1.1.19. ■

Lemma 1.1.21 (Ken Brown's Lemma). *Let $\mathcal{C}$ be a model category and $\mathcal{D}$ a homotopic category². Let*

²A category which has a specified class of weak equivalences which satisfy the 2-out-of-3 axiom. This differs from Riehl's definition, but in particular the 2-out-of-6 axiom implies the 2-out-of-3 axiom, so Ken Brown's lemma applies

$F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor which takes trivial cofibrations between cofibrant objects to weak equivalences. Then F takes all weak equivalences between cofibrant objects to weak equivalences. Dually, if F takes trivial fibrations between fibrant objects to weak equivalences, then F takes all weak equivalences between fibrant objects to weak equivalences.

Proof. Let $f : a \rightarrow b$ is a weak equivalence between cofibrant objects. The map $(f, 1_b) : a \amalg b \rightarrow b$ factors via (α, β) into a cofibration $q : a \amalg b \rightarrow c$ and a trivial fibration $p : c \rightarrow b$. Since 0 is initial,

$$\begin{array}{ccc} 0 & \longrightarrow & a \\ \downarrow & & \downarrow \\ b & \longrightarrow & a \amalg b \end{array}$$

is a pushout. Since a and b are cofibrant objects, corollary 1.1.20 shows that the inclusions $i_a : a \rightarrow a \amalg b$ and $i_b : b \rightarrow a \amalg b$ are cofibrations. Note that by definition of the coproduct, $pqi_a = f$ and $pqi_b = 1_b$. Since p is also a weak equivalence, since it is a trivial fibration, we have by the 2-out-of-3 axiom that qi_a and qi_b are weak equivalences, and thus trivial cofibrations. In particular, they are trivial cofibrations $b \rightarrow c$ and $a \rightarrow c$. Note also that the composition

$$0 \longrightarrow a \xrightarrow{i_a} a \amalg b \xrightarrow{q} c$$

is a composition of cofibrations, and thus c is cofibrant. Therefore, qi_a and qi_b are trivial cofibrations between cofibrant objects. By hypothesis, $F(qi_a)$ and $F(qi_b)$ are weak equivalences. Since $F(pqi_b) = F(1_b)$ is also a weak equivalence, by the 2-out-of-3 property $F(p)$ is a weak equivalence. By the 2-out-of-3 property, $F(p \circ q \circ i_1) = F(f)$ is a weak equivalence. ■

Intuition 1.1.22. Ken Brown's lemma says if a functor preserves trivial cofibrations between cofibrant objects, then we can expand it to all weak equivalences, and vice versa for fibrants.

1.2 The homotopy category

In this section, we follow Quillen's original approach with few modifications to construct the homotopy category $\text{Ho } \mathcal{C}$ by formally inverting all weak equivalences in $\mathcal{C}$. The main result is that the quotient category $\mathcal{C}_{cf}/\sim$ obtained by quotienting cofibrant and fibrant objects by the homotopy relation is equivalent as a category to $\text{Ho } \mathcal{C}$. We will see what this means soon.

Definition 1.2.1 (Homotopy Category). Let $\mathcal{C}$ be a homotopic category with weak equivalences $\mathcal{W}$. The *homotopy category* $\text{Ho } \mathcal{C}$ is given as follows: Take the free category $F(\mathcal{C}, \mathcal{W}^{-1})$ on $\mathcal{C}$ and $\mathcal{W}^{-1}$, where the latter is the formal inversion of $\mathcal{W}$. Objects are objects of $\mathcal{C}$, and morphisms are chains of composable pairs $(f_1, \dots, f_n)$ where each f_i is either an arrow in $\mathcal{C}$ or the formal reversal w_i^{-1} of an arrow in $\mathcal{W}$. Composition is string concatenation, and the identities are empty strings. Quotient by the relations $1_a = (1_a)$ for all $a \in \mathcal{C}$, $(f, g) = (g \circ f)$ for all composable pairs (f, g) , and $1_{\text{dom } w} = (w, w^{-1})$ and $1_{\text{cod } w} = (w^{-1}, w)$ for all $w \in \mathcal{W}$. The resultant category is $\text{Ho } \mathcal{C}$. ♥

Intuition 1.2.2. The way to view these relations is that we may freely remove identities from strings without changing their value, we may freely compose arrows without changing their value, and this composition assigns for each $w \in \mathcal{W}$ an inverse w^{-1} .

to her notion of a homotopic category as well.

Remark 1.2.3. The category $\mathrm{Ho} \mathcal{C}$ is also sometimes denoted by $\mathcal{C}[\mathcal{W}^{-1}]$, since it is the localization of $\mathcal{C}$ at $\mathcal{W}$. Also note that hom-sets may not in general be sets in $\mathrm{Ho} \mathcal{C}$, so we must implicitly pass to a higher universe in general. We will show soon that this is not necessary in the context of model categories. It is clear from the definition that $\mathrm{Ho} D\mathcal{C} = (\mathrm{Ho} \mathcal{C})^{\mathrm{op}}$ if $\mathcal{C}$ is a model category. We can also show that given model categories $\mathcal{C}$ and $\mathcal{D}$, $\mathrm{Ho}(\mathcal{C} \times \mathcal{D}) \cong \mathrm{Ho} \mathcal{C} \times \mathrm{Ho} \mathcal{D}$.

One notes that we should intuitively think of $\mathrm{Ho} \mathcal{C}$ as the “quotient” by the relation that maps in $\mathcal{W}$ be inverted, and thus we would expect it to be characterized by the universal property of quotients; namely that there is a localization functor $\gamma : \mathcal{C} \rightarrow \mathrm{Ho} \mathcal{C}$ which is the identity on objects and $f \mapsto [f]$ on morphisms, which should be initial with respect to functors that invert weak equivalences. Indeed, such a functor clearly exists, and the following lemma verifies the universal property.

Lemma 1.2.4. *Suppose $\mathcal{C}$ is any category with a subcategory $\mathcal{W}$.*

1. *If $F : \mathcal{C} \rightarrow \mathcal{D}$ is a functor that sends maps of $\mathcal{W}$ to isomorphisms, then there is a unique functor $\mathrm{Ho} F : \mathrm{Ho} \mathcal{C} \rightarrow \mathcal{D}$ such that $(\mathrm{Ho} F) \circ \gamma = F$;*
2. *Suppose $\delta : \mathcal{C} \rightarrow \mathcal{E}$ is a functor that takes maps of $\mathcal{W}$ to isomorphisms and satisfies the universal property of (1). Then there is a unique isomorphism $F : \mathrm{Ho} \mathcal{C} \rightarrow \mathcal{E}$ such that $F\gamma = \delta$;*
3. *The correspondence of part (1) induces an isomorphism of categories between the functor category $\mathcal{D}^{\mathrm{Ho} \mathcal{C}}$ and the subcategory of functors $\mathcal{D}^{\mathcal{C}}$ which invert isomorphisms of $\mathcal{W}$.*

Proof. One could skip most of this proof and simply appeal to the fact that this is a special case of a quotient category, for which we already have these results, but in order to be thorough I work through the proof without appealing to this fact.

Define $\mathrm{Ho} F$ to be F on objects, and $\mathrm{Ho} F(w^{-1}) = F(w)^{-1}$. We also define $\mathrm{Ho} F$ to be F on arrows not in $\mathcal{W}^{-1}$; an inspection of the relations we are quotienting by shows that this is well-defined:

- we quotient by the relation that we may compose arrows. This is an axiom of categories, and thus all functors necessarily identify these arrows simply by the fact that their domain is a category;
- we quotient by the relation that composing by 1 doesn’t change the value of a composition. Once again, this is an axiom of categories themselves;
- we quotient by the relation that arrows in $\mathcal{W}$ have inverses, which $\mathrm{Ho} F$ respects.

Functoriality follows since F is functorial: $\mathrm{Ho} F(gf) = F(gf) = F(g)F(f)$ provided $g, f \notin \mathcal{W}^{-1}$; if $u, v \in \mathcal{W}$ then define $w = uv$, and we have $w^{-1} = v^{-1}u^{-1} = (uv)^{-1}$. We obtain

$$\mathrm{Ho} F(v^{-1}u^{-1}) = \mathrm{Ho} F(w^{-1}) = F(w)^{-1};$$

$$\mathrm{Ho} F(v^{-1}) \mathrm{Ho} F(u^{-1}) = F(v)^{-1}F(u)^{-1} = (F(u)F(v))^{-1} = F(uv)^{-1} = F(w)^{-1}.$$

Thus, our definition applies on $\mathcal{W}^{-1}$ as well. A similar proof takes care of the case where only one arrow is in $\mathcal{W}^{-1}$. We then have

$$(\mathrm{Ho} F \circ \gamma)(x) = \mathrm{Ho} F(x) = F(x), \quad (\mathrm{Ho} F \circ \gamma)(f) = \mathrm{Ho} F(f) = F(f),$$

and thus $\text{Ho } F \circ \gamma = F$. To show uniqueness, let $\text{Ho } F \circ \gamma = F$; we must show the definition of $\text{Ho } F$ is forced. Indeed, $\text{Ho } F$ must be F on objects and arrows, so we only need to show our definition on $\mathcal{W}^{-1}$ is forced. Indeed, this definition is forced by functoriality and basic properties of isomorphisms, since $\text{Ho } F(w^{-1})$ must be the unique two-sided inverse of $\text{Ho } F(w) = F(w)$, and thus $\text{Ho } F(w^{-1}) = F(w)^{-1}$. This shows property (1).

Now for property (2). Let $\delta : \mathcal{C} \rightarrow \mathcal{E}$ also be universal with respect to inverting $\mathcal{W}$. This proof follows in the standard way: by part (1) there is a unique functor $F : \text{Ho } \mathcal{C} \rightarrow \mathcal{E}$ such that $F\gamma = \delta$, and since δ is also universal, by (1) there is a unique functor $G : \mathcal{E} \rightarrow \text{Ho } \mathcal{C}$ such that $G\delta = \gamma$. Left composing by G on our first relation gives

$$F\gamma = \delta \implies GF\gamma = G\delta.$$

Since $G\delta = \gamma$, we have that $GF\gamma = \gamma$. Similarly, left composing our second relation by F gives $FG\delta = \delta$. Finally, note that the identity $1_{\text{Ho } \mathcal{C}} : \text{Ho } \mathcal{C} \rightarrow \text{Ho } \mathcal{C}$ is the unique arrow induced by universality of γ with respect to itself^a; in other words $1_{\text{Ho } \mathcal{C}}$ is the unique arrow such that $1_{\text{Ho } \mathcal{C}}\gamma = \gamma$. Since we also have that $GF\gamma = \gamma$, we conclude that $GF = 1_{\text{Ho } \mathcal{C}}$. The exact same proof for the other relation shows that $FG = 1_{\mathcal{E}}$, and thus F and G are isomorphisms.

Finally, define the isomorphism by $F \mapsto \text{Ho } F$. This is obviously well-defined, but since the definition of $\text{Ho } F$ is uniquely forced by F , it has a well-defined inverse $\text{Ho } F \mapsto F$ given simply by precomposition by γ . Now let $\tau : F \Rightarrow G$; we must define $\text{Ho } \tau : \text{Ho } F \Rightarrow \text{Ho } G$. We may freely define it componentwise as $\text{Ho } \tau_x = \tau_x$, since the target categories are the same in both functor categories. We need only show $\text{Ho } \tau_x$ is natural with respect to inverses weak equivalences, since τ is natural on F and G , and $\text{Ho } F, \text{Ho } G$ agree with F, G outside of weak equivalences. This is easiest seen algebraically:

$$\tau_y F(w) = G(w) \tau_x \implies \tau_y = G(w) \tau_x F(w)^{-1} \implies G(w)^{-1} \tau_y = \tau_x F(w)^{-1} \implies \text{Ho } G(w)^{-1} \tau_y = \tau_x \text{Ho } F(w)^{-1}.$$

Vertical composition of natural transformations is preserved manifestly, and horizontal composition is a nonfactor since this is a functor category. ■

^aSince γ inverts weak equivalences, by universality of γ there is an arrow $\text{Ho } \mathcal{C} \rightarrow \text{Ho } \mathcal{C}$ as defined in part (1).

Remark 1.2.5. After working through the algebra for (3) and recalling a proof I did in enriched categories, I am beginning to think that if one has an isomorphism, commutivity of any diagram is preserved by swapping an arrow with its isomorphism. Although I don't see how to prove this in the general case, and as such it may not hold in the general case, I can see an obvious proof if the isomorphism is on the “end” of the diagram, meaning there are either no arrows postcomposing with it, or no arrows precomposing with it. One then simply notes that if they wrote out the algebraic expressions the diagram represents, the isomorphism is always either on the front or back of the equation, and one can simply compose on the relevant side with the inverse.

Proposition 1.2.6. *Let $\mathcal{C}$ be a model category, and write $\mathcal{C}_c$, $\mathcal{C}_f$, and $\mathcal{C}_{cf}$ for the full subcategories of cofibrant, fibrant, and cofibrant and fibrant objects of $\mathcal{C}$. Then the inclusion functors induce equivalences of categories $\text{Ho } \mathcal{C}_{cf} \rightarrow \text{Ho } \mathcal{C}_c \rightarrow \text{Ho } \mathcal{C}$ and $\text{Ho } \mathcal{C}_{cf} \rightarrow \text{Ho } \mathcal{C}_f \rightarrow \text{Ho } \mathcal{C}$.*

Proof. The referenced functors are induced as follows: the localizations in the diagram

$$\begin{array}{ccccc} \mathcal{C}_{cf} & \longrightarrow & \mathcal{C}_c & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Ho} \mathcal{C}_{cf} & & \mathrm{Ho} \mathcal{C}_c & & \mathrm{Ho} \mathcal{C} \end{array}$$

invert weak equivalences, and thus the composition of a localization with the inclusion inverts weak equivalences. Universality as in lemma 1.2.4 induces functors

$$\begin{array}{ccccc} \mathcal{C}_{cf} & \longrightarrow & \mathcal{C}_c & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Ho} \mathcal{C}_{cf} & \dashrightarrow & \mathrm{Ho} \mathcal{C}_c & \dashrightarrow & \mathrm{Ho} \mathcal{C} \end{array}$$

as required. We first show the inverse of the inclusion $i : \mathcal{C}_c \hookrightarrow \mathcal{C}$ is given by cofibrant replacement. We have $(Q \circ i)(x) = Q(x)$, and since $Q(x)$ is cofibrant, this is a valid functor to $\mathcal{C}_c$. Since x is also guaranteed to be cofibrant since $x \in \mathcal{C}_c$, and since $\mathcal{C}_c$ is full, the component $q_x = \beta(0 \rightarrow x) : Q(x) \rightarrow x$ as defined in remark 1.1.16 exists in $\mathcal{C}_c$ for each cofibrant x . Since $Qi(x) = Q(x)$, by remark 1.1.16, this gives a natural transformation $q : Qi \rightarrow 1_{\mathcal{C}_c}$. Moreover, this transformation is pointwise trivial fibrant. The functors lower to

$$\mathrm{Ho} \mathcal{C}_c \xrightleftharpoons[\mathrm{Ho} Q]{\mathrm{Ho} i} \mathrm{Ho} \mathcal{C}$$

and q lowers to $\mathrm{Ho} q : \mathrm{Ho} Q \circ \mathrm{Ho} i \Rightarrow 1_{\mathrm{Ho} \mathcal{C}_c}$ by lemma 1.2.4. Since $\mathrm{Ho} q_x = q_x$ and q_x is a trivial fibration, and in particular is a weak equivalence in $\mathcal{C}_c$, we have that q_x is an isomorphism in $\mathrm{Ho} \mathcal{C}_c$ since localization inverts weak equivalences. Thus, since each component of q_x is an isomorphism, we have a natural isomorphism $\mathrm{Ho} q : \mathrm{Ho} Q \circ \mathrm{Ho} i \cong 1_{\mathrm{Ho} \mathcal{C}_c} = \mathrm{Ho} 1_{\mathcal{C}_c}$. The converse direction is the same, since $iQ = Q$, so it follows directly by this argument and remark 1.1.16.

Now we show the inclusion $i : \mathcal{C}_{cf} \hookrightarrow \mathcal{C}_c$ induces an equivalence of categories. The inverse will in this case be given by fibrant replacement and cofibrant replacement. Note that each $x \in \mathcal{C}_{cf}$ is cofibrant and fibrant by assumption. We have that $(R \circ i)(x) = R(x)$ is fibrant, but we must also show it is cofibrant. Note that x is cofibrant, so the arrow $0 \rightarrow x$ is a cofibration. By the axioms of a model category, the arrow $\gamma(x \rightarrow *) : x \rightarrow R(x)$ is a trivial cofibration, and in particular a cofibration. Composing these gives an arrow $0 \rightarrow R(x)$ which is a cofibration, and thus $R(x)$ is cofibrant. Now note that there is a natural trivial cofibration $r : 1_{\mathcal{C}} \Rightarrow R$ by remark 1.1.16. Since $r_x : x \rightarrow R(x)$ has fibrant domain and codomain, it exists in $\mathcal{C}_{cf}$, and thus we obtain a natural trivial cofibration $r : 1_{\mathcal{C}_{cf}} \Rightarrow Ri$. By the same logic as above, localization inverts r into a natural isomorphism. The converse direction is once again the same, since once again $iR = R$. ■

Remark 1.2.7. An interesting result from my argument is that $R(x)$ preserves cofibrancy, and dually $Q(x)$ preserves fibrancy. In other words, if x is fibrant, then so is $Q(x)$, and if x is cofibrant, then so is $R(x)$.

We thus have that $\mathrm{Ho} \mathcal{C}_{cf} \simeq \mathrm{Ho} \mathcal{C}$. There is an alternative construction of $\mathrm{Ho} \mathcal{C}_{cf}$ for model categories for which hom-sets are indeed sets, which we now summarize. We denote by $\Delta : x \rightarrow x \times x$ the diagonal map and $\nabla : x \amalg x \rightarrow x$ the dual map, called the *fold map*.

Definition 1.2.8 (Homotopy). Let $\mathcal{C}$ be a model category, and let $f, g : b \rightarrow x$ be maps in $\mathcal{C}$.

1. A *cylinder object* for b is a factorization of the fold map $\nabla : b \amalg b \rightarrow b$ into a cofibration $b \amalg b \xrightarrow{i_0+i_1} b'$ followed by a weak equivalence $b' \xrightarrow{s} b$;
2. A *path object* for x is a factorization of the diagonal map $\Delta : x \rightarrow x \times x$ into a weak equivalence $x \xrightarrow{r} x'$ followed by a fibration $x' \xrightarrow{(p_0, p_1)} x \times x$;
3. A *left homotopy* from f to g is a map $H : b' \rightarrow x$ for some cylinder object b' for b such that $H i_0 = f$ and $H i_1 = g$. We say that f and g are *left homotopic*, written $f \sim^\ell g$, if there is a left homotopy from f to g ;
4. A *right homotopy* from f to g is a map $K : b \rightarrow x'$ for x' some path object for x , such that $p_0 K = f$ and $p_1 K = g$. We say that f and g are *right homotopic*, written $f \sim^r g$, if there is a right homotopy from f to g ;
5. We say that f and g are *homotopic*, written $f \sim g$, if they are both left and right homotopic;
6. f is a *homotopy equivalence* when there is some $h : x \rightarrow b$ such that $hf \sim 1_b$ and $fh \sim 1_x$.

♥

Intuition 1.2.9. The reason we refer to cylinder objects as cylinders is most naturally seen after developing intuition for a left homotopy. Diagrammatically, a left homotopy H from f to g is a commutative diagram

$$\begin{array}{ccccc}
 b & & & & \\
 \downarrow & \searrow^{i_0} & & \searrow^{f} & \\
 b \amalg b & \xrightarrow{i_0+i_1} & b' & \xrightarrow{H} & x \\
 \uparrow & \nearrow_{i_1} & & \nearrow_{g} & \\
 b & & & &
 \end{array}$$

where $i_0 + i_1$ is a cofibration, and b' is weakly equivalent to b as noted above. Ignore for a moment the cofibration and weak equivalence semantics. Imagine b' as an actual cylinder $B \times I$ for B a topological space and I the unit interval. If $f, g : B \rightarrow X$ are continuous maps, a regular homotopy between them is a continuous map $H : B \times I \rightarrow X$ which agrees with F on $B \times \{0\} \cong B$ and agrees with g on $B \times \{1\} \cong B$. Although these subspaces are isomorphic, they are disjoint as subsets of the cylinder $B \times I$. Thus the intuition is that b in the diagram is the domain space of the morphisms, and b' is the cylinder $b \times I$. The inclusion i_0 inserts b into the bottom face of the cylinder at 0, and i_1 includes it at the top face at 1. A homotopy is intuitively thought of as a map along the entire cylinder to x , agreeing with f and g on the given faces, and this is indeed what the diagram above expresses. Of course there is not necessarily a unit interval object I in $\mathcal{C}$, so this is a further generalization of this notion. The weak equivalence puts a constraint on how far this generalization goes. In topology, the cylinder is homotopy equivalent to any of its faces by contracting along I ; in particular it is homotopy equivalent to b . The weak equivalence $b' \rightarrow b$ enforces this rule in model categories as well. Finally, I am unsure why $i_0 + i_1$ need be a cofibration, but my intuition is starting to be that cofibrations are an axiomatic generalization of monomorphisms, in which case this makes total sense as well. We will have to see going forward if this intuition holds.

Once again, we begin by considering the diagrammatic representation of a right homotopy,

before discussing the term *path object*. A right homotopy from f to g is a diagram

$$\begin{array}{ccccc}
 & & f & \xrightarrow{\quad} & x \\
 & \nearrow & & \nearrow p_0 & \uparrow \\
 b & \xrightarrow{H} & x' & \xrightarrow{(p_0, p_1)} & x \times x \\
 & \searrow & & \searrow p_1 & \downarrow \\
 & & g & \xrightarrow{\quad} & x
 \end{array}$$

where (p_0, p_1) is a fibration, and x' is weakly equivalent to x as noted above. Note the obvious fact that path objects are dual to cylinder objects, and right homotopies are dual to left homotopies. Once again, we imagine B, X as topological spaces. We will see that this notion of homotopy is less familiar from topology, and more familiar from simplicial homotopy in a cotensored simplicial category. The path object in this case should be thought of as the path space X^I . The maps p_0 and p_1 are then thought of as evaluating paths in X^I at 0, and evaluating paths in X^I at 1. The map (p_0, p_1) is then just $\gamma \mapsto (\gamma(0), \gamma(1))$. A homotopy is then a continuous map $B \rightarrow X^I$, such that evaluating at 0 agrees with f and evaluating at 1 agrees with g . It can be thought of as encoding the information of f and g into the path space by evaluating along the unit interval. The weak equivalence once again guarantees the homotopy equivalence $X^I \sim X$, which follows since evaluation is continuous, and $X^* \cong X$ for any singleton $* \subseteq I$. The fact that (p_0, p_1) is a fibration implies that one may lift homotopies along the map from $X \times X$ to X^I .

Remark 1.2.10. By duality, left homotopies in $\mathcal{C}$ are right homotopies in $D\mathcal{C}$ and vice versa, and the same with path and cylinder objects. Thus, one need only prove statements about left homotopies and cylinder objects in order to prove them for path objects and right homotopies.

Remark 1.2.11. One immediately considers the applications of the functorial factorizations to the construction of path objects and cylinder objects. Indeed, apply the functorial factorization (α, β) to $\nabla : b \amalg b \rightarrow b$ to obtain a cylinder object $Q(b \amalg b)$, with a cofibration $b \amalg b \rightarrow Q(b \amalg b)$ and a trivial fibration $Q(b \amalg b) \rightarrow b$. This factorization has the added bonus of the weak equivalence being a fibration, and the cylinder object being cofibrant. The book denotes this object by $b \times I$, which is purely notation and does not indicate an actual product. In some enriched categorical settings, this can however be literal instead of notation, such as topological or simplicial categories. Dually, we obtain a path object $R(x)$ for x by the other factorization, which has the benefit of being fibrant, and the weak equivalence being a cofibration. We will sometimes denote this object as x^I when viewing it as a path object.

Proposition 1.2.12. *Let $\mathcal{C}$ be a model category, and $f, g : b \rightarrow x$ maps in $\mathcal{C}$.*

1. *If $f \sim^\ell g$ and $h : x \rightarrow y$, then $hf \sim^\ell hg$. Dually, if $f \sim^r g$ and $h : a \rightarrow b$, then $fh \sim^r gh$;*
2. *If x is fibrant, $f \sim^\ell g$, and $h : a \rightarrow b$, then $fh \sim^\ell gh$. Dually, if b is cofibrant, $f \sim^r g$, and $h : x \rightarrow y$, then $hf \sim^r hg$;*
3. *If b is cofibrant, then left homotopy is an equivalence relation on $\mathcal{C}(b, x)$. Dually, if x is fibrant, then right homotopy is an equivalence relation on $\mathcal{C}(b, x)$;*
4. *If b is cofibrant and $h : x \rightarrow y$ is a trivial fibration or a weak equivalence of fibrant objects, then h induces an isomorphism*

$$\mathcal{C}(b, x) / \sim^\ell \cong \mathcal{C}(b, y) / \sim^\ell.$$

Dually, if x is fibrant and $h : a \rightarrow b$ is a trivial cofibration or a weak equivalence of cofibrant

objects, then h induces an isomorphism

$$\mathcal{C}(b, x)/\sim^r \cong \mathcal{C}(a, x)/\sim^r;$$

5. If b is cofibrant, then $f \sim^\ell g$ implies $f \sim^r g$. Furthermore, if x' is any path object for x , then there is a right homotopy $K : b \rightarrow x'$ from f to g . Dually, if x is fibrant, then $f \sim^r g$ implies $f \sim^\ell g$, and there is a left homotopy from f to g using any cylinder object for b .

Proof. By remark 1.2.10, we need only prove the statements for left homotopies and cylinder objects. The statement of (1) is fairly obvious, given a homotopy $f \sim^\ell g$, consider the diagram

$$\begin{array}{ccccc} b & & & & \\ \downarrow & \searrow^{i_0} & & \searrow^{f} & \\ b \amalg b & \xrightarrow{i_0 + i_1} & b' & \xrightarrow{H} & x \xrightarrow{h} y \\ \uparrow & \nearrow_{i_1} & & \nearrow_{g} & \\ b & & & & \end{array}$$

One immediately sees that composing along h gives hH a homotopy from hf to hg .

To prove (2), let

$$b \amalg b \xrightarrow{i} b' \xrightarrow{s} b$$

be the cylinder object. Note that s is a weak equivalence, and s admits functorial factorizations into a cofibration and a trivial fibration, or a trivial cofibration and a fibration. Note that for either choice, we have that one of the components of the factorization is a weak equivalence, and since the composition is a weak equivalence, by the 2-out-of-3 property we have that the third map is a weak equivalence. Thus s factors as a trivial cofibration followed by a trivial fibration

$$b' \xrightarrow{k} b'' \xrightarrow{s'} b.$$

Since $k(i_0 + i_1)$ is a composition of cofibrations, we may consider b'' to be a cylinder object for b . Because x is fibrant, the homotopy $H : b' \rightarrow x$ extends to a homotopy $H : b'' \rightarrow x$. We may in particular use the factorization (γ, δ) to choose b'' to be fibrant by remark 1.1.16, giving a diagram

$$\begin{array}{ccccc} & & s & & \\ & \nearrow^k & & \searrow_{s'} & \\ b' & \xrightarrow{\quad} & b'' & \xrightarrow{\quad} & b \\ \downarrow H & & \downarrow & & \\ x & & & & \\ \downarrow & & & & \\ * & & & & \end{array}$$

where I have denoted fibrations by $\twoheadrightarrow$ and cofibrations by $\rightarrowtail$. Note that the triangle commutes by universality of $*$, and we may rewrite this commutative triangle as the commutative square of

solid arrows

$$\begin{array}{ccc} b' & \xrightarrow{H} & x \\ k \downarrow & \nearrow H' & \downarrow \\ b'' & \twoheadrightarrow & 1 \end{array}$$

Since k is a trivial cofibration and $x \rightarrow *$ a fibration, and since trivial cofibrations have the left lifting property with respect to all fibrations, the dotted arrow as indicated exists, rendering the diagram commutative. Since $k(i_0 + i_1)$ and b'' form a cylinder object as noted above, and $H'k = H$, we have that the homotopy H does indeed extend to a homotopy H' for f and g . With this understanding, we may without loss of generality take the arrow from the cylinder object to b to be a trivial fibration.

Quickly note that $h\nabla_a = \nabla_b(h \amalg h)$, which is an identity that holds in generality. We see this by comparing the two diagrams

$$\begin{array}{ccc} a & & \\ \downarrow & \searrow 1 & \\ a \amalg a & \xrightarrow{\nabla_a} & a \\ \uparrow & \nearrow 1 & \\ a & & \end{array} \quad \xrightarrow{h} \quad \begin{array}{ccccc} a & \xrightarrow{h} & b & & \\ \downarrow & & \downarrow & \searrow 1 & \\ a \amalg a & \xrightarrow{h \amalg h} & b \amalg b & \xrightarrow{\nabla_b} & b \\ \uparrow & & \uparrow & \nearrow 1 & \\ a & \xrightarrow{h} & b & & \end{array}$$

and noting that composing along the h gives the same arrows $a \rightarrow b$ in both diagrams, so by universality the compositions $a \amalg a \rightarrow b$ must be the same. Noting this fact, now let

$$a \amalg a \xrightarrow{j} a' \xrightarrow{t} a$$

be a cylinder object for a , which exists by remark 1.2.11. We note that the diagram

$$\begin{array}{ccccc} a \amalg a & \xrightarrow{h \amalg h} & b \amalg b & \xrightarrow{i_0 + i_1} & b' \\ j \downarrow & \searrow \nabla_a & \searrow \nabla_b & & \downarrow s \\ a' & \xrightarrow{t} & a & \xrightarrow{h} & b \end{array}$$

commutes, since the definition of a cylinder object guarantees $s(i_0 + i_1) = \nabla_b$ and $tj = \nabla_a$, hence the dotted lines, and the middle square expresses the equality $\nabla_b(h \amalg h) = h\nabla_a$. Thus we obtain that the diagram of solid lines

$$\begin{array}{ccc} a \amalg a & \xrightarrow{(i_0 + i_1)(h \amalg h)} & b' \\ j \downarrow & \nearrow u & \downarrow s \\ a' & \xrightarrow{ht} & b \end{array}$$

commutes. Since j is cofibrant by definition of a cylinder object, and s is fibrant by our earlier assumption, we obtain a lift, indicated by the dotted line, rendering the diagram commutative.

Let H be the homotopy from f to g . We claim that Hu is a homotopy fh to fg . Indeed, since

$$\begin{array}{ccccc}
 a & \xrightarrow{h} & b & \xrightarrow{f} & x \\
 \downarrow & & \downarrow & \searrow i_0 & \uparrow \\
 a \amalg a & \xrightarrow{h \amalg h} & b \amalg b & \xrightarrow{i_0 + i_1} & b' \xrightarrow{H} x \\
 \uparrow & & \uparrow & \nearrow i_1 & \uparrow \\
 a & \xrightarrow{h} & b & \xrightarrow{g} & x
 \end{array}$$

commutes, and since $uj = (i_0 + i_1)(h \amalg h)$ by the lifting diagram, we replace these arrows and obtain that the diagram

$$\begin{array}{ccccccc}
 a & \xrightarrow{h} & b & \xrightarrow{f} & & & x \\
 \downarrow & & \downarrow & \searrow i_0 & & & \uparrow \\
 a \amalg a & \xrightarrow{j} & a' & \xrightarrow{u} & b' & \xrightarrow{H} & x \\
 \uparrow & & \uparrow & \nearrow i_1 & & & \uparrow \\
 a & \xrightarrow{h} & b & \xrightarrow{g} & & & x
 \end{array}$$

commutes, where the arrows $a \rightarrow a'$ arise by universality, and thus uH is indeed a left homotopy $fh \sim^\ell gh$, as desired.

Now consider (3). Note that $\sim^\ell$ is obviously symmetric, since we need only flip the diagram horizontally. Showing $\sim^\ell$ is reflexive is also simple: by corollary 1.2.13, there is a cylinder object $\nabla = s \circ (i_0 + i_1)$ given by a functorial factorization, and postcomposing by f gives a diagram

$$\begin{array}{ccccc}
 b & \xrightarrow{f} & x \\
 \downarrow & \searrow i_0 & \uparrow \\
 b \amalg b & \xrightarrow{i_0 + i_1} & b' \xrightarrow{s} b \xrightarrow{f} x \\
 \uparrow & \nearrow i_1 & \uparrow \\
 b & \xrightarrow{f} & x
 \end{array}$$

and thus fs is a homotopy $f \sim^\ell f$. To show it's transitive, we must now make use of the hypothesis that b be cofibrant. Let $H : b' \rightarrow x$ be a homotopy from f to g , and let $H' : b'' \rightarrow x$ be a homotopy from g to h . Let $i_0 + i_1$ be the relevant map corresponding to H , and $i'_0 + i'_1$ the map corresponding to H' . Define c via the pushout square

$$\begin{array}{ccc}
 b & \xrightarrow{i_1} & b' \\
 i'_0 \downarrow & & \downarrow \\
 b'' & \longrightarrow & c
 \end{array}$$

and define j_0, j_1 as the compositions

$$\begin{aligned} j_0 : b &\xrightarrow{i_0} b' \longrightarrow c, \\ j_1 : b &\xrightarrow{i'_1} b'' \longrightarrow c. \end{aligned}$$

We want $j_0 + j_1 : b \amalg b \rightarrow c$ to define a cylinder object, but we need the first map to be a cofibration, and we need a weak equivalence $t : c \rightarrow b$. We will focus on the second one for now. Define t as the map induced by universality of pushout in the diagram

$$\begin{array}{ccc} b & \xrightarrow{i_1} & b' \\ i'_0 \downarrow & & \downarrow \\ b'' & \longrightarrow & c \end{array} \quad \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \\ \xrightarrow{s'} \end{array} \quad \begin{array}{c} \\ \\ b \end{array}$$

and note that b is cofibrant. Note that if we can show i_1 is a trivial cofibration, then by corollary 1.1.20 so is $b'' \rightarrow c$, and since $t \circ (b'' \rightarrow c) = s'$ and s' is a weak equivalence, we have that by the 2-of-3 property t is a weak equivalence.

Note that by universality of 0 , the pushout of $b \leftarrow 0 \rightarrow b$ is the coproduct

$$\begin{array}{ccc} 0 & \longrightarrow & b \\ \downarrow & & \downarrow \\ b & \longrightarrow & b \amalg b \end{array}$$

and by corollary 1.1.20 the inclusions into the coproduct are cofibrations. Since $i_1 = (i_0 + i_1) \circ \iota_1$ for $\iota_1 : b \rightarrow b \amalg b$ the relevant inclusion is a composition of cofibrations, we have that i_1 is a cofibration. By remark 1.1.19, all isomorphisms are trivial cofibrations; in particular the identity. Since $s \circ i_1 = 1_b$, and since s and 1_b are weak equivalences, by 2-out-of-3 so is i_1 .

Note here that

$$b \amalg b \xrightarrow{j_0 + j_1} c \xrightarrow{t} b$$

is a factorization of the fold map, since j_0 factors along si_0 :

$$\begin{array}{ccccc} b & & & & \\ \downarrow & \searrow i_0 & & \searrow j_0 & \\ b \amalg b & \xrightarrow{i_0 + i_1} & b' & \longrightarrow & c \xrightarrow{t} b \\ \uparrow & \nearrow i_1 & & \nearrow s & \\ b & & & & \end{array}$$

commutes by inspection of the pullback diagram and the definitions of j_0 and t . The same can be said for j_1 . Thus we obtain $tj_0 = si_0 = 1_b$, and the same for j_1 . We would have our cylinder

now, except for the fact that $j_0 + j_1$ need not be a cofibration. Before we fix this, note that the homotopies H and H' induce a map $K : c \rightarrow x$ along the pullback:

$$\begin{array}{ccc}
 b & \xrightarrow{i_1} & b' \\
 i'_0 \downarrow & & \downarrow \\
 b'' & \longrightarrow & c \\
 & \searrow & \downarrow K \\
 & & x
 \end{array}
 \begin{array}{l}
 \text{curved arrow } H \text{ from } b' \text{ to } x \\
 \text{curved arrow } H' \text{ from } b'' \text{ to } x
 \end{array}$$

Note that once again by inspection of the pullback diagram and by the definitions of j_0 and j_1 , the diagrams

$$\begin{array}{ccccc}
 f : b & \xrightarrow{i_0} & b' & \xrightarrow{\quad} & c \xrightarrow{K} x \\
 & \searrow j_0 & \nearrow & & \uparrow H \\
 & & & & c \\
 h : b & \xrightarrow{i'_1} & b'' & \xrightarrow{\quad} & c \xrightarrow{K} x \\
 & \searrow j_1 & \nearrow & & \uparrow H \\
 & & & & c
 \end{array}$$

commute, and thus we have $Kj_0 = f$ and $Kj_1 = h$. One would hope that we can turn K into a homotopy by finding a cofibration to replace $j_0 + j_1$, and indeed we can. Factor $j_0 + j_1$ into a cofibration and a trivial fibration $b \amalg b \rightarrow c' \rightarrow c$. We can show that c' is a cylinder object for b , and that the composition

$$c' \longrightarrow c \xrightarrow{K} x$$

is a left homotopy between f and h . Indeed, since the arrow $c' \rightarrow c$ is a *trivial* fibration, it is in particular a weak equivalence, and thus composing with t gives another weak equivalence. Thus we have a weak equivalence $c' \rightarrow b$. Denote by z the cofibration $b \amalg b \rightarrow c'$. We then have a diagram

$$\begin{array}{ccccc}
 b & & & & \\
 \downarrow \iota_0 & \searrow z\iota_0 & & & \\
 b \amalg b & \xrightarrow{z} & c' & \longrightarrow & c \xrightarrow{K} x \\
 \uparrow \iota_1 & \nearrow z\iota_1 & & & \\
 b & & & &
 \end{array}
 \begin{array}{l}
 \text{curved arrow } j_0 \text{ from } b \text{ to } c \\
 \text{curved arrow } j_1 \text{ from } b \text{ to } c
 \end{array}$$

which commutes, where I have labeled the inclusions simply for clarity. Since all we did was factor $j_0 + j_1$, we still have that the composition is the fold map, and thus we do indeed have the stated homotopy, as required.

Next, we show (4). We start with the trivial fibration case. Let $h : x \rightarrow y$ be a trivial fibration, and define the map

$$F : \mathcal{C}(b, x) / \sim^\ell \rightarrow \mathcal{C}(b, y) / \sim^\ell$$

by postcomposition $F[f] = [hf]$, which is well-defined by (1) and (3). We first show F is surjective. Let $f' : b \rightarrow y$ be some map. Since b is cofibrant and h is a trivial fibration, the lifting problem

indicated by the diagram of solid arrows

$$\begin{array}{ccc}
 0 & \longrightarrow & x \\
 \downarrow & \nearrow f & \downarrow h \\
 b & \xrightarrow{f'} & y
 \end{array}$$

admits a dotted arrow rendering the diagram commutative. We thus have that there exists $f : b \rightarrow x$ such that $hf = f'$, and the operation is surjective without even passing to homotopy classes. Now let $H : b' \rightarrow y$ be a left homotopy $hf \sim^\ell hg$. Thus we have a diagram

$$\begin{array}{ccccc}
 b & \xrightarrow{f} & x & & \\
 \downarrow & \searrow i_0 & \downarrow h & & \\
 b \amalg b & \xrightarrow{i_0+i_1} & b' & \xrightarrow{H} & y \\
 \uparrow & \nearrow i_1 & \uparrow h & & \\
 b & \xrightarrow{g} & x & &
 \end{array}$$

and one notes that $hf + hg = h(f + g)$ by universality, and $hf + hg = H(i_0 + i_1)$ also by universality. One thus constructs the diagram of solid arrows

$$\begin{array}{ccc}
 b \amalg b & \xrightarrow{f+g} & x \\
 \downarrow i_0+i_1 & \nearrow K & \downarrow h \\
 b' & \xrightarrow{H} & y
 \end{array}$$

and since h is a trivial fibration and $i_0 + i_1$ is a cofibration, there is a lift indicated by the dotted arrow rendering the diagram commutative. Note that the lift K is indeed homotopy from f to g by universality of coproduct, and thus F is injective. Therefore, F is indeed an isomorphism.

Now we must consider the case where $h : x \rightarrow y$ is a weak equivalence between fibrant objects. One notes that we might hope to view h as a natural transformation $h : \mathcal{C}(-, x)/\sim^\ell \Rightarrow \mathcal{C}(-, y)/\sim^\ell$. By assumption, the domain and codomain are fibrant objects, and thus they exist in $\mathcal{C}_f$. By (1) and (3), the quotient operation is compatible with composition. Thus we have a functor

$$x \xrightarrow{y} \mathcal{C}_{cf}(-, x) \longmapsto \mathcal{C}_{cf}(-, x)/\sim^\ell$$

given by taking the Yoneda embedding of $\mathcal{C}_f$, restricting the resultant functors to $\mathcal{C}_{cf}$ to ensure the domain is cofibrant as well, then composing with the quotient map. The notation $\mathcal{C}_{cf}(-, x)$ is thus an abuse of notation, since x need not be cofibrant; we simply write this to indicate the arguments the functor can take must be fibrant and cofibrant. By the first case, if $h : x \rightarrow y$ is a trivial fibration, then if we can show it is natural, there will be an induced isomorphism $\mathcal{C}_{cf}(-, x) \cong \mathcal{C}_{cf}(-, y)$. By taking weak equivalences in the target category to be isomorphisms, we have that trivial fibrations, in particular trivial fibrations between fibrant objects, are taken to

weak equivalences. By Ken Brown's lemma 1.1.21, this expands to all weak equivalences, proving our statement. It thus suffices to show the given definition of h is natural. Indeed, we have that for any $f : b \rightarrow c$,

$$\begin{array}{ccc} \mathcal{C}(c, x)/\sim^\ell & \xrightarrow{h_*} & \mathcal{C}(c, y)/\sim^\ell \\ f^* \downarrow & & \downarrow f^* \\ \mathcal{C}(b, x)/\sim^\ell & \xrightarrow{h_*} & \mathcal{C}(b, y)/\sim^\ell \end{array}$$

commutes simply by definition of precomposition and postcomposition. Thus, the statement is proven.

Finally, we show (5). Suppose b is cofibrant and $f \sim^\ell g$ by a homotopy $H : b' \rightarrow x$. We must show $f \sim^r g$ using any path object. Since b is cofibrant, the maps i_0, i_1 are trivial cofibrations by the same argument used in part (3) with the pushout of $0 \rightarrow b$. Let

$$x \xrightarrow{r} x' \xrightarrow{(p_0, p_1)} x \times x$$

be a path object for x . Note that since $Hi_0 = f$ and $si_0 = 1_b$, the diagram

$$\begin{array}{ccc} & & b \\ & \nearrow s & \downarrow f \\ b & \xrightarrow{i_0} & b' \\ & \searrow H & \downarrow \\ & & x \end{array}$$

commutes. By universality of product, one concludes that $(fs, H)i_0 = (fsi_0, Hi_0) = (f, f)$ by this diagram. Similarly, note that since $p_0r = p_1r = 1_x$, we have that

$$\begin{array}{ccccc} & & x & & \\ & & \uparrow p_0 & & \\ b & \xrightarrow{f} & x & \xrightarrow{r} & x' \\ & & \downarrow p_1 & & \\ & & x & & \end{array}$$

commutes, and by universality once again we have that $(p_0, p_1)rf = (f, f)$. We conclude that the diagram of solid arrows

$$\begin{array}{ccc} b & \xrightarrow{rf} & x' \\ i_0 \downarrow & \nearrow J & \downarrow (p_0, p_1) \\ b' & \xrightarrow{(fs, H)} & x \times x \end{array}$$

commutes. Since i_0 is trivial cofibrant and (p_0, p_1) is fibrant, there exists a lift as indicated by the dotted arrow rendering the diagram commutative. Finally, we show that $K = Ji_1$ is a right homotopy from f to g . Indeed, by the above diagram

$$(p_0, p_1)Ji_1 = (p_0, p_1)(fs, H)i_1 = (p_0f si_1, p_1Hi_1) = (p_0f, p_1g)$$

since $Hi_1 = g$ and $si_1 = 1$. Thus the restriction along p_0 is f and the restriction along p_1 is g , and Ji_1 is a homotopy, as required. ■

Corollary 1.2.13. *Let $\mathcal{C}$ be a model category. If b is cofibrant and x is fibrant, then the left and right homotopy relations coincide, and are equivalence relations on $\mathcal{C}(b, x)$. Moreover, if $f \sim g : b \rightarrow x$, then there is a left homotopy $H : b' \rightarrow x$ from f to g using any cylinder object b' for b . Dually, there is a right homotopy $k : b \rightarrow x'$ from f to h using any path object x' for b .*

Proof. Indeed, that the relations coincide follows by remark 1.2.11 (5), the fact that they are equivalence relations follows from (4), and the invariance under cylinder/path objects is (5) once again. ■

Corollary 1.2.14. *The homotopy relation on the morphisms of $\mathcal{C}_{cf}$ is an equivalence relation and is compatible with composition. Hence the category $\mathcal{C}_{cf}/\sim$ exists.*

Remark 1.2.15. From our proof to remark 1.2.11 (3), we saw that if b is cofibrant, then the inclusions $b \rightarrow b \amalg b$ are cofibrations, and thus the arrows $i_0, i_1 : b \rightarrow b'$ are trivial cofibrations. Dually, the arrows $p_0, p_1 : x' \rightarrow x$ are trivial fibrations when x is fibrant.

Recall our result from ?? that $\text{Ho } \mathcal{C}_{cf} \simeq \text{Ho } \mathcal{C}$. It would definitely be nice to show that the quotient category by the homotopy relation is equivalent to $\text{Ho } \mathcal{C}_{cf}$, and thus to $\text{Ho } \mathcal{C}$, since this branch of math is literally called *homotopy theory*. Our first step in proving this would be to show that $\mathcal{C}_{cf}/\sim$ inverts weak equivalences, thus at least getting us a functor $\text{Ho } \mathcal{C}_{cf} \rightarrow \mathcal{C}_{cf}/\sim$ by universality of localization. We accomplish this in the following proposition.

Proposition 1.2.16. *Let $\mathcal{C}$ be a model category. Then a map in $\mathcal{C}_{cf}$ is a weak equivalence if and only if it is a homotopy equivalence. In particular, quotienting by homotopy inverts weak equivalences.*

Proof. First let $f : a \rightarrow b$ be a weak equivalence of cofibrant and fibrant objects. By ??, we have a natural isomorphism $f_* : (\mathcal{C}_{cf}/\sim)(x, a) \rightarrow (\mathcal{C}_{cf}/\sim)(x, b)$ for any fibrant and cofibrant x . Choose $x = b$; there then necessarily exists a map $g : b \rightarrow a$ which maps by postcomposition by f to 1_b ; hence $fg \sim 1_b$. By ??, we then have $fgf \sim f$, so taking $x = a$ now gives $fgf \sim f$ in the codomain, and thus there is some k such that postcomposition $fk \sim fgf \sim f$. Necessarily by injectivity, k admits a definition as both gf and 1 , and thus $gf \sim 1$. Thus, f is a homotopy equivalence.

The converse is more involved. Let f be a homotopy equivalence between cofibrant and fibrant objects. Factor f into a trivial cofibration $g : a \rightarrow c$ and a fibration $p : c \rightarrow b$. Clearly c is cofibrant, since the unique arrow from $0 \rightarrow c$ factors as $g \circ (0 \rightarrow a)$, which is a composition of cofibrations. Similarly, $(b \rightarrow *) \circ p$ is a composition of fibrations, so c is cofibrant and fibrant. By the proof of the other direction, since g is a trivial cofibration and in particular a weak equivalence, it is a homotopy equivalence. By the 2-out-of-3 property, it suffices to show p is a weak equivalence. Let $f' : b \rightarrow a$ be a homotopy inverse to f , and let $H : b' \rightarrow b$ be the left homotopy $ff' \sim 1_b$. Since

$f = pg$, one factors $f f' = pg f' = H i_0$, and constructs the commutative diagram of solid arrows:

$$\begin{array}{ccc} b & \xrightarrow{gf'} & c \\ i_0 \downarrow & \nearrow H' & \downarrow p \\ b' & \xrightarrow{H} & b \end{array}$$

Since b is cofibrant, by remark 1.2.15 i_0 is a trivial cofibration, and since p is a fibration, there is a dotted arrow as indicated rendering the diagram commutative. Define $q = H' i_1$, and note that $p H' = H$ implies that $p q = 1_b$, since H is a homotopy $f f' \sim 1_b$. Note also that since $H' i_0 = g f'$, we have that H' is a homotopy from $g f'$ to q . Since we found earlier that g was a homotopy equivalence, let g' be a homotopy inverse for g . Then $p \sim p g g' \sim f g'$. We thus conclude that

$$q p \sim (g f')(f g') \sim 1_c.$$

Let $K : c' \rightarrow c$ be a left homotopy from 1_c to $q p$, and note that $K i_0 = 1_c$ is a weak equivalence since it is an isomorphism. By remark 1.2.15, i_0 is a weak equivalence, and by the 2-out-of-3 property, so is K . Since i_1 is a weak equivalence by the same remark, we thus have that $K i_1 = q p$ is also a weak equivalence. Recalling that $p q = 1_b$, we have that the diagram

$$\begin{array}{ccccc} c & \xrightarrow{1} & c & \xrightarrow{1} & c \\ p \downarrow & & q p \downarrow & & \downarrow p \\ b & \xrightarrow{q} & c & \xrightarrow{p} & b \end{array}$$

commutes. We also have that the top and bottom rows are the identity, and thus p is a retract of $q p$. By the retract axiom, p is a weak equivalence. ■

Corollary 1.2.17. *Let $\mathcal{C}$ be a model category, and let $\gamma : \mathcal{C}_{cf} \rightarrow \text{Ho } \mathcal{C}_{cf}$ and $\delta : \mathcal{C}_{cf} \rightarrow \mathcal{C}_{cf}/\sim$ be the canonical functors. Then there is a unique isomorphism $j : \mathcal{C}_{cf}/\sim \cong \text{Ho } \mathcal{C}_{cf}$ such that $j \delta = \gamma$. In particular, j is the identity on objects.*

Proof. By universality, it suffices to show $\mathcal{C}_{cf}/\sim$ has the same universal property as described in lemma 1.2.4. We know already that δ inverts weak equivalences by proposition 1.2.16. Let $F : \mathcal{C}_{cf} \rightarrow \mathcal{D}$ be a functor which inverts weak equivalences. Let

$$a \amalg a \xrightarrow{i_0 + i_1} a' \xrightarrow{s} a$$

be a cylinder object for a . Then $s i_0 = s i_1 = 1_a$ by assumption. Since s is a weak equivalence, and F inverts weak equivalences,

$$F(1_a) = F(s)F(i_0) = F(s)F(i_1) \implies F(s)^{-1}F(s)F(i_0) = F(i_0) = F(i_1) = F(s)^{-1}F(s)F(i_1).$$

Thus, if $H : a' \rightarrow b$ is a left homotopy between $f, g : a \rightarrow b$, then

$$F(f) = F(H)F(i_0) = F(H)F(i_1) = F(g).$$

Thus, F identifies homotopic maps. By universality of quotient categories, there is a unique functor $\mathcal{C}_{cf}/\sim \rightarrow \mathcal{D}$ as required.

Note that we did not show a' was cofibrant and fibrant. We need not show this for $a \amalg a$, since we did not compute a functor on an arrow with $a \amalg a$ as domain or codomain. We define a' by a functorial factorization, and by remark 1.2.11 it does not matter which cylinder object we choose, so this is fine. By remark 1.2.15 the arrow $i_0 : a \rightarrow a'$ is a cofibration, and since $0 \rightarrow a \rightarrow a'$ is a composition of cofibrations, a' is cofibrant. Since a is fibrant, and $a' \rightarrow a$ is a fibration, we have that a' is fibrant as well, concluding the proof. ■

Theorem 1.2.18. *Let $\mathcal{C}$ be a model category. Let $\gamma : \mathcal{C} \rightarrow \mathrm{Ho} \mathcal{C}$ be the localization functor, and let Q and R denote the cofibrant and fibrant replacements of $\mathcal{C}$, respectively, as defined in remark 1.1.16.*

1. *The inclusion $\mathcal{C}_{cf} \rightarrow \mathcal{C}$ induces an equivalence of categories*

$$\mathcal{C}_{cf}/\sim \cong \mathrm{Ho} \mathcal{C}_{cf} \cong \mathrm{Ho} \mathcal{C};$$

2. *There are natural isomorphisms*

$$\mathcal{C}(QR(x), QR(y))/\sim \cong \mathrm{Ho} \mathcal{C}(\gamma(x), \gamma(y)) \cong \mathcal{C}(RQ(x), RQ(y))/\sim.$$

In addition, there is a natural isomorphism

$$\mathrm{Ho} \mathcal{C}(\gamma(x), \gamma(y)) \cong \mathcal{C}(Q(x), R(y))/\sim,$$

and if x is cofibrant and y is fibrant, there is a natural isomorphism

$$\mathrm{Ho} \mathcal{C}(\gamma(x), \gamma(y)) \cong \mathcal{C}(x, y).$$

In particular, $\mathrm{Ho} \mathcal{C}$ is a category without passing to a higher universe;

3. *The functor $\gamma : \mathcal{C} \rightarrow \mathrm{Ho} \mathcal{C}$ identifies left and right homotopic maps;*
4. *If $f : a \rightarrow b$ is a map in $\mathcal{C}$ such that $\gamma(f)$ is an isomorphism in $\mathrm{Ho} \mathcal{C}$, then f is a weak equivalence.*

Proof. (1) follows by corollary 1.2.17 and ???. Investigation of the inverses as noted in ??? gives the natural isomorphisms of (2). The rest of (2) follows from the natural weak equivalences $Q \Rightarrow 1 \Rightarrow R$ as noted in remark 1.1.16, and invoking remark 1.2.11 (3,5). Part (3) is subsumed by corollary 1.2.17. For (4), let $f : a \rightarrow b$ be a map in $\mathcal{C}$ which is inverted by γ . Then $QR(f)$ is an isomorphism in $\mathcal{C}_{cf}/\sim$, and thus $QR(f)$ is necessarily a homotopy equivalence. By corollary 1.2.17, $QR(f)$ is a weak equivalence. Since $Q \Rightarrow 1$ is a natural weak equivalence,

$$\begin{array}{ccc} QR(a) & \xrightarrow{q_{R(a)}} & R(a) \\ QR(f) \downarrow & & \downarrow R(f) \\ QR(b) & \xrightarrow{q_{R(b)}} & R(b) \end{array}$$

commutes, and in particular $QR(f)$ and $q_{R(b)}$ are weak equivalences. Thus $q_{R(b)}QR(f) = R(f)q_{R(a)}$ is a weak equivalence, and since $q_{R(a)}$ is also a weak equivalence, so is $R(f)$. The same argument with $1 \Rightarrow R$ shows f is a weak equivalence. ■

Remark 1.2.19. We now have a variety of remarkable facts. Localization inverts precisely those maps that are weak equivalences; Riehl calls categories for which this is true *saturated*. We also have that weak equivalences are precisely homotopy equivalences, which is a remarkable fact and implies that model categories provide excellent structure for homotopy theory in a general setting. We have shown that homotopy categories of model categories are locally small, and that we can even choose to work in the category given by quotienting by homotopy, if we are willing to wrangle the inverse maps when necessary. We also have that localization identifies homotopic maps, another excellent cue that model categories are an excellent setting for homotopy theory. Finally, the book notes it will denote $\mathrm{Ho}\mathcal{C}(x, y)$ by $[x, y]$ as we continue through the text. I will likely adopt this convention as well since $\mathrm{Ho}\mathcal{C}(x, y)$ is quite annoying to type, but I don't like the square brackets much either, and really think we should come up with a new notation.